The youngest layered rocks (sequence A) are older than 636 Ma and comprise polygenic cobble- and boulder-conglomerate containing granite and metavolcanic clasts, lithic and arkosic arenite, red siltstone and limestone, deposited in a shallow marine environment.

All three sequences are intruded by 640—580 Ma old Pan-African granite and quartz monzonite. Widespread K—Ar cooling ages of ca. 600 Ma indicate regional uplift towards the end of the Precambrian. The rocks of the Bahah province therefore depict a 300 Ma evolution from a primitive island arc to a mature island arc setting. An essentially continental crustal character was achieved in the very late Precambrian following the emplacement of Pan-African granites.

GEOLOGY AND MINERALIZATION AT MUSAYNA'AH AREA, SAUDI ARABIA

T. KADI, J. TAYEB, A. AL-SHANTI and ABDEL AZIZ HUSSEIN

Faculty of Earth Sciences, King Abdulaziz University, Jeddah (Saudi Arabia)

The Musayna'ah area is built up of Precambrian rocks of the Upper Afna and Nuqrah Formations of the Hulayfah Group, and the lower part of the Ar-Ragab Formation of the Murdama Group. The dominant types are volcaniclastic tuffs, basic, intermediate and acidic volcanics and marble, all intruded by granite, granodiorite, diorite and dykes of various composition.

Mineralization occurs at two prospects, namely, Musayna'ah and Jabal Al-Hamra. The ore minerals at Musayna'ah are pyrite, chalcopyrite and magnetite, while those at Jabal Al-Hamra are magnetite, hematite and some pyrite.

It is here suggested that the mineralization at Musayna'ah is in the nature of 'epithermal vein-type deposits' characteristic of the andesite—rhyolite volcanic piles in subaerial or shallow water environments in continental margins above subduction zones.

The magnetite—hematite deposit at Jabal Al-Hamra was formed at the contact between granodiorite and andesitic tuff intercalated with carbonate bands, as a result of a convection cell, whereby iron and copper were leached from the tuffs by healed groundwater at the hot end of the convective cell and deposited within the carbonate bands at the colder end of the cycle.

No economic significance is attached to either prospect.

PALEOMAGNETIC INVESTIGATION OF UPPER PROTEROZOIC ROCKS IN THE EASTERN ARABIAN SHIELD, KINGDOM OF SAUDI ARABIA

KARL S. KELLOGG

U.S. Geological Survey Mission, Jeddah (Saudi Arabia)

GEORGE E.J. BECKMANN

King Abdulaziz University, Jeddah (Saudi Arabia)

Paleomagnetic investigations of samples from 42 sites in the vicinity of Bi'r Juqjuq (latitude 21° 20′ N, longitude 43° 40′ E) in the southeastern Arabian shield suggests that elevated crustal temperatures ca. 620—600 Ma ago may have magnetically-reset many of

the older rocks. Samples include andesitic volcanic rocks and hematized metasedimentary rocks tentatively assigned to the late Proterozoic Halaban Group (ca. 760—700 Ma old), a post-Halaban andesite porphyry stock, a post-Halaban granodiorite batholith (ca. 620 Ma old), and two post-Halaban felsic dikes. Samples from many sites demonstrate unstable or scattered paleomagnetic directions based on alternating-field demagnetization to 1000 Oe and thermal demagnetization to 680° C. However, stable paleomagnetic directions from two sites (12 samples) in the granodiorite and from five sites (19 samples) in the Halaban volcanic rocks are similar and yield a mean direction of magnetization of D = 351.8°, I = 30.9° (downward), α_{95} = 8.8°. These directions define a paleomagnetic pole at latitude 81.0° N., longitude 284.1° E. (dp = 5.5°, dm = 9.8°). The authors suggest that the stable magnetization measured in rocks of the Halaban Group was imposed during regional cooling shortly after emplacement of the granodiorite pluton about 620 Ma ago. A similar mean paleomagnetic pole (latitude 75.5° N, longitude 325.6° E) determined from 22 magnetically-stable samples from the two felsic dikes indicates that the dikes may be contemporaneous with, or slightly younger than, the granodiorite. A baked-contact test on one of the dike samples demonstrates that the dike possesses original thermal remanent magnetization. After a correction for the opening of the Red Sea in Tertiary times, the two calculated pole positions are close to the pole established for Gondwanaland at 600 Ma ago. This correspondence supports the contention that this part of the Arabian craton was joined to the African craton by that time.

A different magnetization direction has been determined from 12 magnetically-stable samples from a hornblende andesite porphyry stock that postdates the granodiorite (D = 289.3°, I = 43.5°, α_{95} = 11.8°). The resulting virtual geomagnetic pole at latitude 25.8° N, longitude 332.2° E (after correcting for Red Sea rifting) best fits the Gondwana polar-wander path at ca. 510 Ma ago.

This investigation suggests that stable magnetization was preserved only after consolidation of the crustal blocks that have been proposed by various workers to comprise the southeastern shield. The authors belive that this consolidation event (continental collision and termination of subduction) and the widespread generation of granitic magmas were responsible for a pervasive thermal pulse that obliterated earlier magnetizations. If these conclusions are applicable to the remainder of the southern Arabian shield, it will not be possible to reconstruct paleomagnetically the relative positions of subelements of the shield prior to ca. 620 Ma ago.

ON THE ANALYSIS OF SIMPLE MACROSCOPIC FOLDS IN THE EASTERN DESERT OF EGYPT — A FIELD-BASED MODEL AT WADI UM LASSIFA

SAMIR M. KHAWASIK

Department of Geology, Faculty of Science, Aswan (Egypt)

Many areas of the Precambrian in Egypt have been previously mapped and described as 'simple homoclines' when they are, in fact, macroscopically folded. The macroscopic fold, in the modern sense of structural geology, is that which cannot be directly seen in the field, and may not be elucidated by the standard mapping methods alone. This type of fold can only be predicted by 'kinematic analysis'. The macroscopic fold is usually large perhaps measuring several kilometers in size. Many other reasons may hinder the field geologist in recognizing such folds, some of which are discussed in the present paper.